



* Practical - 12 *

* activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.  
com/apk/res/android"
```

```
    android:orientation="vertical"
```

```
    android:padding="16dp"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent" >
```

```
    <TextView
```

```
        android:text="Index Messages:"
```

```
        android:textSize="18sp"
```

```
        android:layout_width="wrap_content" />
```

```
    <ListView
```

```
        android:id="@+id/listViewMessages"
```

```
        android:layout_width="match_parent"
```

```
        android:layout_height="300dp"
```

```
        android:dividerHeight="10dp" />
```

```
    <EditText
```

```
        android:id="@+id/editTextNumber"
```

```
        android:hint="Enter phone number"
```

```
        android:inputType="phone"
```

```
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content" />
```

```
<EditText  
    android:id="@+id/editTextMessage"  
    android:hint="Enter message"  
    android:layout_width="match-parent"  
    android:layout_height="wrap-content"/>
```

```
<Button  
    android:id="@+id/btnSend"  
    android:text="Send SMS"  
    android:layout_width="wrap-content"  
    android:layout_height="wrap-content"  
    android:layout_marginTop="10dp"/>
```

```
</LinearLayout>
```

* MainActivity.java

```
package com.example.myapplication;
```

```
public class MainActivity extends AppCompatActivity {  
    private ListView listViewMessages;  
    private EditText editTextNumber, editTextMessage;  
    private Button btnSend;
```

```
    private static final int REQUEST_SMS_PERMISSION=123;
```

@Override

```
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);
```




```

listViewMessages = findViewById(R.id.listViewMessages);
editTextNumber = findViewById(R.id.editTextNumber);
editTextMessage = findViewById(R.id.editTextMessage);
btnSend = findViewById(R.id.btnSend);

```

```

if (ContextCompat.checkSelfPermission(this, Manifest.
    permission.READ_SMS) != PackageManager.
    PERMISSION_GRANTED || ContextCompat.checkSelfPermission(
    this, Manifest.permission.SEND_SMS) !=
    PackageManager.PERMISSION_GRANTED) {
    ActivityCompat.requestPermissions(this,
        new String[]{Manifest.permission.READ_SMS,
            Manifest.permission.SEND_SMS},
        REQUEST_SMS_PERMISSION);
} else {
    loadInboxMessage();
}

```

```

btnSend.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        sendSmsMessage();
    }
});
}

```

```

private void loadInboxMessages() {
    ArrayList<String> smsList = new ArrayList<>();
    Uri uriSms = Uri.parse("content://sms/inbox");
    ContentResolver cr = getContentResolver();
}

```



```
Cursor cursor = cn.query(Curisms, null,  
null, null, null);
```

```
if (Cursor != null && cursor.moveToFirst()) {  
do {
```

```
String address = cursor.getString(  
cursor.getColumnIndexOrThrow("address"));  
String body = cursor.getString(cursor.getColumnIndexOrThrow("body"));  
smsList.add("From: " + address + "\nMessage: "  
+ body);
```

```
} while (cursor.moveToNext());  
cursor.close();
```

```
}
```

```
ArrayAdapter<String> adapter = new ArrayAdapter<>  
(this, android.R.layout.simple_list_item_1, smsList);  
ListViewMessages.setAdapter(adapter);
```

```
}
```

```
private void SendSmsMessage() {
```

```
String phone = editTextNumber.getText().toString().  
trim();
```

```
String message = editTextMessage.getText().toString().  
trim();
```

```
if (phone.isEmpty() || message.isEmpty()) {
```

```
Toast.makeText(this, "Please enter number  
and message",
```

```
Toast.LENGTH_SHORT).show();
```

```
return;
```

```
}
```




try {

SmsManager smsManager = SmsManager.getDefault();
smsManager.sendTextMessage (C phone, null,
message, null, null);

Toast.makeText (this, "SMS Sent!",

Toast.LENGTH_SHORT).show();

editTextMessage.setText (" ");

} catch (Exception e) {

Toast.makeText (this, "Failed to send SMS: "
+ e.getMessage(), Toast.LENGTH
LONG).show();

}

}

⑨ Override

public void onRequestPermissionsResult (int requestCode,

String[] permissions, int[] grantResults) {

if (requestCode == REQUEST_SMS_PERMISSION) {

if (grantResults.length > 0 && grantResults[0]

== PackageManager.PERMISSION_GRANTED) {

loadInboxMessages();

} else {

Toast.makeText (this, "Permission Denied",

Toast.LENGTH_SHORT).show();

}

}

}

}



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* Output *

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